

12

L2 processing

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The literature on bilingual processing has veritably exploded in the last ten or fifteen years, as there is a lot of interest in finding new ways of addressing the fundamental issue of L2 acquisition: are the processes happening in the mind/brain of L2 speakers qualitatively different or qualitatively the same as the processes that native speakers engage in? A separate but related question is whether it is possible for adult L2 learners to acquire an L2 to nativelike levels. We have been interested in finding answers to these questions from the beginning of this textbook. In this chapter, we turn away from underlying grammatical representations and concentrate instead on how L2 speakers use these linguistic representations to parse language in real time. There is an interesting interplay between representation and processing, however. We have assumed in this book that linguistic representations are created in the course of language acquisition and they are subsequently accessed during language use. But we cannot “see” the linguistic representations themselves (yet), so we have to extrapolate what they might be, based on experimental findings of language use. If scholars find that L2 language

processing employs the same eye-tracking signatures, shows the same ERP reflexes and patterns of processing over time as L1 language processing, it is possible to conclude that L1 and L2 representations are indeed similar. If, on the other hand, we uncover qualitative differences, they may be due to different representations, but also to similar representations accessed differently in processing. The answers to these questions are not settled yet, and they propel some cutting-edge current research. Before we go on to look at some findings, let us review the experimental methodologies we have been mentioning so far that are used to explore language processing.

12.1 Experimental techniques employed in bilingual processing studies

Behavioral measures of sentence processing frequently involve placing participants in front of a computer screen and presenting them with sentences that they read for comprehension or translation.¹ Sometimes, speakers are asked to perform other tasks on top of reading for comprehension, which are supposed to strain their working memory resources in order to see what effect this would have on performance. One commonly used experimental paradigm is the noncumulative moving window technique (also known as self-paced reading), as the reader will recall from Chapter 3. For example, in the experiment reported by Roberts and Felser (2011), sentences and comprehension questions were presented one word at a time, with participants bringing up each subsequent word by pushing a button on the button box. The final word of each sentence was indicated by a full stop. The authors created baseline and experimental items that differed minimally by one word. If there was a difference in the reading times of some region, it had to be due to the experimental manipulation.

- (1) Experimental item: While the band played the song pleased all the customers.
- (2) Baseline item: While the band played the beer pleased all the customers.

¹ For a recent review of bilingual processing using online behavioral measures, see Roberts (2013).

As the reader can appreciate, the initial words “while the band played the song” can immediately be integrated into a transitive structure where *the song* is the object of *played*. However, when the subsequent word *pleased* appears, it shows that the preceding analysis is wrong and a reanalysis is needed. Obviously, this reanalysis takes time, reflected in longer reaction times for incorporating that segment into the preceding sentence structure. Looking at example (2) now, as a band cannot play beer, it is immediately obvious that the verb *play* is used intransitively. Hence speakers do not spend so much time integrating the following words into the structure. The structures that can be studied with this and similar experimental techniques include subject and object relative clauses, high and low relative clause attachment, passive constructions, and others.

Another experimental technique that we introduced in Chapter 3 records the electric activity of the brain cortex collecting electroencephalography (EEG) data, and computing event-related brain potentials (ERPs) relative to the onset of the linguistic event, such as presentation of a phoneme, word, or morpheme. Activation is measured by placing a number of electrodes on a person’s scalp. The biggest advantage of this method is that ERPs provide brain wave measures reflecting the neural activity without delay, with a temporal resolution of less than a millisecond, across the whole utterance (e.g., for each word, syllable, phoneme). Another advantage of ERPs is that they can distinguish between different cognitive processes without requiring metalinguistic tasks.²

One of the most remarkably consistent and replicable results over thirty years of cross-linguistic language-related ERP research is that lexico-semantic and morphosyntactic manipulations elicit qualitatively different brain responses. That, of course, is a nice confirmation of the fundamental linguistic division between lexical and functional morphemes. Now we know that the brain doesn’t treat these two types of morphemes in the same way.

All content words elicit negative voltage with a peak at around 400 ms after presentation of the word, hence *the N400* name (see Figure 12.1).³ The

² See Steinhauer (2014) for a review of this technique and findings from the perspective of the Critical Period Hypothesis.

³ Negative values are plotted upwards on graphs (which does look strange). Note also that N400 can be elicited by nonlinguistic stimuli, such as different colors, or different images.

EEG → Event-related brain potentials (ERPs)

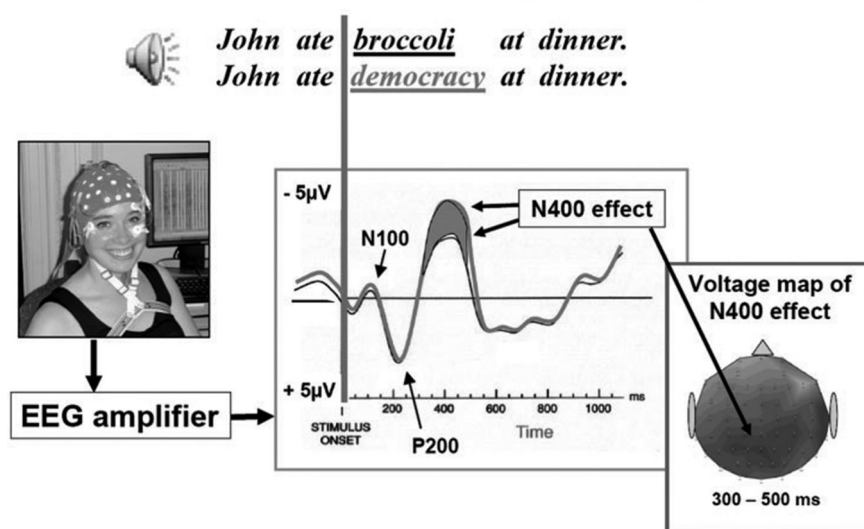


Figure 12.1 EEG setup, N400 effect, and voltage map. EEG data collected from an electrode-cap during sentence processing are averaged time-locked to the target words (*broccoli* and *democracy*). The (schematic) ERP plot shows similar N100s and P200s in both conditions and an enhanced N400 for the semantic mismatch condition (negative polarity is plotted upwards). The voltage map illustrates the centro-parietal scalp distribution of this N400 effect between 300 and 500 ms (mismatch minus control, from Steinhauer (2014))

size of this peak depends on numerous factors, such as a word's semantic relatedness to a preceding context, its probability in the context, and frequency (Kutas and Federmeier 2011, Kutas and Hillyard 1980, Osterhout and Nicol 1999). In the original publication, Kutas and Hillyard (1980), sentences as in (3) were compared. The interesting reaction is to the final noun in bold:

- (3) a. I shaved off my moustache and **beard**. (high probability word)
 b. I shaved off my moustache and **eyebrows**. (low probability word)
 c. I shaved of my moustache and **city**. (incongruent word)

On average, both unexpected final nouns (as in 3b,c) elicited an ERP waveform that was somehow different from the congruent control nouns (3a), but the negativity was larger and more robust for the incongruent word (3c). In later findings, it was argued that the N400 effect indexes not only

semantic integration of the word into the previous sentence and discourse but also the reader/listener's expectations about the word, including those based on real-world knowledge.

Another very important language-related ERP effect is the P600, so called because it is a positive deflection in electrical brain activity starting at 500 milliseconds and peaking at 600 milliseconds after the onset of the triggering word. The P600 is elicited by hearing or reading morphological errors, syntactic reanalysis such as garden path sentences (as in (2)), and other syntactic anomalies (Hagoort et al. 1993). In particular, it has been related to ungrammatical continuations of the preceding material as in (4).

(4) The spoiled child throw the toys on the floor.

As with the N400, it has recently been acknowledged that the P600 may be an umbrella effect for linguistically different processes triggered by ungrammaticality, garden paths, and long-distance *wh*-questions (Gouvea et al. 2010). The LAN (left anterior negativity) is another effect sensitive to morphosyntactic violations and grammatical processes that are at the phrase and clause level, so it comes earlier than the P600 (see Figure 12.2). It is generally considered that the P600 is more reliable while the LAN shows more variability. To reiterate, none of these components are unique to language processing: they have been attested in looking at pictures and listening to music.

The third technology that is used in contemporary psycholinguistic research on bilingualism is informally known as eye tracking.⁴ This technique also measures real-time moment-by-moment processing. Its major advantage is that it captures processing decisions during natural, uninterrupted comprehension without the need to ask the participants for explicit linguistic judgments (Rayner 1998, 2009). The eye-movement tracking method comes in two main varieties. In one, the eyes of an individual are tracked while she is reading a written text. In the other, the eye movements are tracked over a visual scene during listening to sentences. The latter is called the “visual world paradigm.” Research has shown that the eyes go to

⁴ See the special issue of *SSLA* 2013 (2) edited by Paula Winke, Aline Godroid, and Susan Gass, especially the article by Roberts and Siyanova-Chanturia, for a review of the technique and findings.

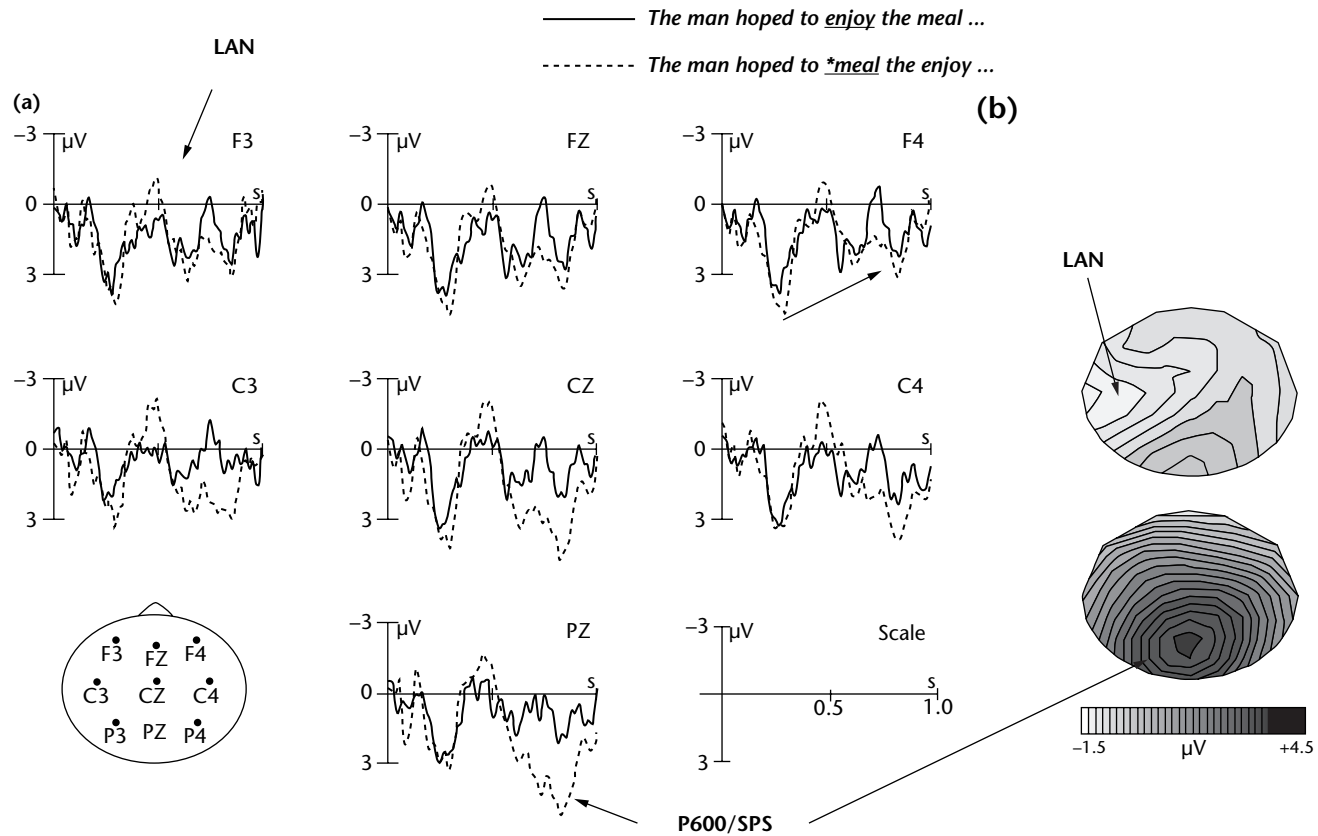


Figure 12.2 Biphasic LAN and P600 effects elicited by a syntactic word category violation in native English readers. (a) ERPs at seven electrodes show a LAN between 300–500 ms and a subsequent posterior P600 (600–1000 ms) for the ungrammatical target word (dotted line). (b) Voltage maps illustrating the scalp distribution of both ERP effects, from Steinhauer (2014)

the image of the word on the screen within 200 ms of the word being pronounced as part of the linguistic signal.

In the former paradigm, eye tracking during reading, rapid eye movements are made from one fixation point to another. These movements are called “saccades” and they are so quick that researchers believe no new information is being input. Between saccades, there are “fixation points,” and these fixation points and the time spent fixed on a word provide valuable information about processing decisions. Function words are not fixated on as long as content words. Although the direction of reading is forward, eye movements go back, too. About 15% of saccades are “regressions” of different duration: to the previous word or to the beginning of the sentence, or even to previous sentences. These regressions are interpreted as signs of processing difficulty. For example, ambiguous words, garden path sentences, or a more unusual integration of a word into a sentence meaning lead to longer regressions. We will discuss results of both types of studies in this chapter. Figure 12.3 from Roberts and Siyanova-Chanturia 2013 presents a hypothetical eye-movement record.

In addition to ERPs, eye-tracking, functional magnetic resonance imaging (fMRI) and magneto-encephalography (MEG) are used to study the neuronal underpinnings of language comprehension and production. The literature of this actively developing field of psycholinguistic inquiry is huge and growing every month. All these new techniques are most effective in elucidating human language processing when they are based on well-understood linguistic distinctions such as the distinction between lexical and functional morphemes or between nouns and verbs. Although there are significant disconnects between linguistic theory and the neurobiology of language (Poeppel and Embick 2005), there are also promising areas of

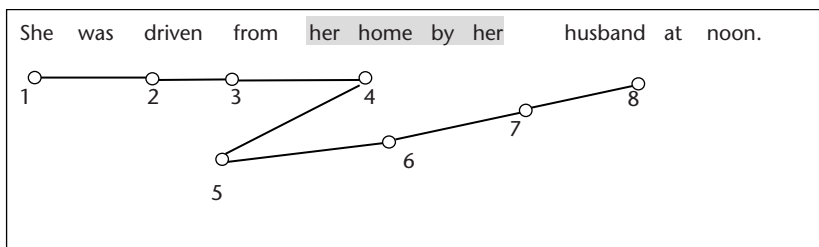


Figure 12.3 Hypothetical eye-movement record, with the gray area representing the region of interest, from Roberts and Siyanova-Chanturria (2013)

Notes: First fixation duration = 3, first pass reading time = 3+4, total reading time = 3+4+6, regression path duration = 3+4+5+6, rereading = 5+6, second pass reading time = 6, fixation count = 3+4+6 (3 fixations). First fixation duration and first pass reading time are considered early measures; the rest are considered late measures.

investigation where the neurosciences intersect with linguistic research more closely than before (Poeppel et al. 2012). Much of the integration between linguistics and neuroscience is in the future.

In the next sections, we will approach the processing of the different parts of the grammar: inflectional morphology, syntax, and lexical and context integration. Keep in mind that not all constructions that we covered in the previous chapters can be evaluated during processing, due to the various constraints of the experimental methodologies. However, there are enough constructions and properties representative of the three areas of the grammar that we can draw conclusions on how these areas are processed in real time and in natural comprehension. I will discuss some theoretical accounts of bilingual processing and what predictions they make for behavior. Then we will square these predictions with the findings we encounter.

12.2 Accounts and predictions

In considering bilingual processing, we will have to keep in mind the theoretical divide with respect to L1–L2 differences. Generally speaking, the two opposing groups of accounts mirror the competence accounts, arguing for fundamental difference or for fundamental similarity. As the reader will remember, among fundamental difference in competence accounts are Bley-Vroman's Fundamental Difference Hypothesis (1989, and especially 2009) and the Representational Deficit view of Tsimpli and Hawkins (Hawkins and Hattori 2006, Tsimpli and Dimitrakopoulou 2007). For processing, models with parallel predictions of difference are Clahsen and Felser's Shallow Structure Hypothesis (2006), and Ullman's declarative/procedural model (2005). These accounts are essentially in keeping with the Critical Period Hypothesis that we discussed in Chapter 4. The common underpinning of all these accounts is that once maturational constraints during early and late childhood have resulted in loss of plasticity in brain systems needed for native language acquisition, late learners have to allocate alternative brain systems for their L2 acquisition and have to approach the learning tasks with qualitative different mechanisms. Because learners' grammatical representations are different, their language processing routines and mechanisms are also qualitatively different from native ones. We shall see more concrete predictions as we go over the different areas of processing.

On the other side of the divide are models and proposals that consider L1 and L2 competence and processing mechanisms to be quantitatively, but not qualitatively different. Such models include the Full Transfer/Full Access/Full Parse model (Dekydtspotter, Schwartz, Sprouse 2006), Hopp's Fundamental Similarity Hypothesis (e.g., 2007) and Steinhauer et al.'s (2009) Convergence Hypothesis. Alan Juffs's seminal work on L2 processing, although it does not provide a separate model, also supports this position (Juffs and Harrington 1996, Juffs 1998a,b). These models share the prediction that with increased levels of proficiency, the signatures of L2 processing of the various parts of the grammar will be approaching or converging with native processing signatures. L1 influence is not excluded by these accounts, but it is deemed possible to overcome at later levels of experience, since the processing mechanisms in L1 and L2 acquisition are essentially the same.

12.3 L2 processing of functional morphology

In Chapter 9, we discussed whether L2 speakers can decompose words into roots, stems, and affixes (derivational and inflectional). Recall that this decomposition works in the following way. Roots, stems, and morphemes are stored in the mental lexicon, together with the rules for their composition. For example *workers* is made up of the root *work*, the derivational affix *-er*, and the inflectional plural morpheme *-s*. *Work-er* is the stem for the addition of the plural morpheme. If a word is very frequent, it is likely that it will be stored whole by native as well as L2 speakers. If a word is less frequent, then composition of the constituent morphemes happens fast and online. At issue is whether L2 speakers are also capable of executing this process, as native speakers do. While results are mixed, we have enough experimental evidence from various highly inflected languages (Russian, Spanish, Serbian) showing that learners of these languages do indeed decompose. However, these studies investigated lexical access in isolation, usually through masked priming tasks.

In this section, we will look at how the inflectional morphology is processed in whole sentences and connected discourse. This is actually quite a different cognitive process compared to mental lexicon access, although the latter is certainly part of it. Functional morphemes have phonological realization; they also represent a bundle of grammatical meanings, or

features that determine their syntactic behavior, i.e., their distribution within words and phrases relative to other morphemes and words in the sentence. In order to incorporate a functional morpheme in a sentence and evaluate its acceptability, it is commonly necessary to look at the whole sentence, not just the containing word. It is exactly this level in second language processing that we will review in this section. In the name of clarity, we will make a slightly artificial division between morpho-syntactic inflections and syntax proper, in the understanding that these two processes are completely integrated in language processing.

Neuroscientific data from the 1990s and early 2000s provided evidence in support of the view that the L2 morphosyntax presents insurmountable difficulties to bilinguals and that they are never up to scratch, compared to native speakers. The most influential study from that time is Weber-Fox and Neville (1996). The researchers replicated parts of the classic Johnson and Newport (1989) study, documenting a fundamental difference between late and early L2 learners, thus significantly boosting the Critical Period Hypothesis. Weber-Fox and Neville investigated various morphosyntactic and semantic violations read by adult Chinese and Korean learners of English. Two representative test items are given in (5) and (6). In the morphosyntactic violations as in (5) the preposition *of* and the noun head *proof* in the NP+PP item were switched. In the semantic violations as in (6), a similar structure with genitive nouns was sought. ERP components in reading acceptable and unacceptable sentences were compared for learners of various ages of arrival in the US, as well as native speakers.

(5) *He criticized Max's of proof the theorem.

(6) *John sailed Mary's cloud to Boston.

Only the very early learners (AoA before 2) showed an early LAN-like effect, as well as an appropriate P600. The adolescent arrivals exhibited a P600 but no LAN. Morphosyntactic violations did not elicit even P600s in the late arrivals. On the other hand, the semantic anomalies as in (6) elicited N400s in all groups, suggesting that grammar is processed differently from meaning. Later experiments (Hahne and Friederici 2001, Hahne 2001) extended these findings for auditory presentation of the stimuli. As Steinhauer (2014: 404) points out, by the year 2001, behavioral and ERP evidence seemed to strongly back up the Critical Period Hypothesis.

However, researchers in this area of psycholinguistics noticed a confound almost immediately. In studies that divide learners in groups according to age of arrival in the US, such as Johnson and Newport (1989) and Weber-Fox and Neville (1996), proficiency is confounded with age of arrival. Typically, a bilingual who arrived in an English-speaking country at the age of 2 is at a higher level of proficiency when tested than a bilingual who arrived in her or his twenties. This of course is a major problem for these studies, because it is not clear whether their results can be attributed to the later age of acquisition or to the lower proficiency of the bilinguals. We have previously discussed that the CPH should be falsified if we find even very few late learners to be capable of nativelike acquisition. This verification mechanism has another twist when processing is concerned. If the strong CPH claim is to be supported with neurophysiological findings, studies must show that successful learners use *different brain mechanisms* than native speakers. We are going to look at some later studies, which document similar brain mechanisms for L1 and L2 processing.

Another problem with the early studies, pointed out by Steinhauer (2014: 404–05) is their unbalanced experimental design. Since ERP effects are calculated only in comparison with the baseline, it is imperative that the baseline and the experimental sentences have exactly the same words before the violation. However, this is not the case for the experimental (5) and the baseline (7).⁵ Can you see what the difference is?

(5) *He criticized Max's **of** proof the theorem.

(7) He criticized Max's proof **of** the theorem.

In general, many ERP researchers have argued that not all morphosyntactic violations reliably elicit LAN-like negativities in native speakers; sometimes native speakers show N400s followed by P600s (e.g., Tanner and van Hell 2014). In such cases, the lack of LANs cannot be considered as an indication of nonnative processing. In addition, ERP components can overlap in time, so that a positivity and a negativity, if they overlap even partially, can cancel themselves (at midline electrodes). Until these methodological issues with the technique are resolved to a general level of satisfaction, the lack of LAN effect in L2 learners should not be capitalized upon.

⁵ Detailed criticism can be found in Steinhauer and Drury (2012).

A number of new studies have appeared, documenting similar ERP effects in natives and bilinguals, thereby casting doubt on the earlier findings (e.g., Rossi et al. 2006). These studies are careful not to confound age of acquisition and proficiency. For example, Rossi et al. (2006) tested Italians who were late learners of German and Germans who were late learners of Italian. This study documented that learners at high proficiency showed the native processing pattern of LAN plus P600 when detecting subject–verb agreement violations. The learners at lower levels of proficiency, matched with the other group on age of arrival, elicited a P600 only.

Gillon Dowens et al. (2010) showed that another interesting factor may be affecting nonnative processing: proximity of the agreeing elements. This study looked at number and gender agreement violations between determiner and noun within the NP, and between noun and adjective used predicatively, at a longer distance, in English–Spanish interlanguage. They found ERP patterns qualitatively similar to native ones, consisting of LAN, P600, and later negativities. However, those patterns were only elicited by the local agreement violations between determiner and noun. Violations of noun–adjective agreement did not elicit an early LAN. Thus, language proficiency alone cannot be responsible for all native–nonnative differences. The distance at which agreement is computed (local versus long-distance) also influences brain responses. L1 transfer of the various properties under investigation is also not ruled out, since behavioral measures show L2 learners of Spanish being less accurate on noun–adjective agreement than on determiner–noun agreement. The conclusion of L1 transfer is clinched by a later study by the same authors, Gillon Dowens et al. (2011), testing proficient Chinese learners of Spanish. Since Chinese has very little inflectional morphology, it was expected that the early processing signatures would not be attested. These expectations were confirmed: the Chinese learners showed P600 effects only.

It is instructive to compare findings on functional morphology from the two methodologies, ERPs and eye-tracking. Keating (2009) is an eye-tracking study that exploits the same property as Gillon Dowens et al. (2010). The critical constructions are as exemplified below in (8), (9), and (10), where the distance between the agreeing elements is manipulated.

(8) In the NP:

Una casa pequeña cuesta mucho en San Francisco.
'A small house costs a lot in San Francisco.'

- (9) In the matrix clause VP:

La casa es bastante pequeña y necesita muchas reparaciones.
 ‘The house is quite small and needs a lot of repairs.’

- (10) In the subordinate clause VP:

Una casa cuesta menos si es pequeña y necesita reparaciones.
 ‘A house costs less if it is small and needs repairs.’

In the eye-tracking data, both longer fixation times on the critical elements (in this case the adjective *pequeña*), as well as regressions from the adjective back to the noun for (literally) checking the agreement indicate sensitivity to this grammatical property as well as processing difficulties. The learners’ behavior showed an effect of proficiency as well as an effect of the distance between the noun head and the adjective. The native speakers displayed a response pattern including fixation and subsequent regression for all three noun–adjective distances. Only the highly advanced learners revealed the same nativelike pattern of behavior, and only for the most local configuration as in (8). However, if we consider the stimuli from the point of view of grammatical representations, the same formal features are involved in the noun–adjective agreement no matter the distance between head and modifier. Therefore, Keating argued that his eye-tracking data suggests successful acquisition but more labored processing. His learners had acquired the feature checking mechanism, but had a harder time applying it at a distance, in cases where they had to keep more words in working memory. Thus, the results of the eye tracking and the ERP studies of the same construction cohere completely.

Briefly comparing Keating (2009) to another eye tracking study, Foucart and Frenck-Mestre (2012), brings forward another reason for performance divergence. In the latter study, the agreeing noun and adjective are separated by the auxiliary *sont* ‘are’:

- (11) Au printemps les pommes [Fem] sont vertes [Fem]/ *verts [Masc] sur cet arbre.

‘In spring, apples are green on this tree.’

While all Keating’s subjects, even the native speakers, did not show longer fixation times for ungrammatical items, Foucart and Frenck-Mestre’s natives and learners did spend a longer time fixated on the ungrammatical

forms. Roberts and Siyanova-Chanturia (2013) attribute this difference to the task that participants had to perform. In Keatings's study, participants performed a meaning related task, while in the Foucart and Frenck-Mestre's study, participants' attention to grammar may have been heightened because they were asked to offer a grammaticality judgment. Thus, the different results could be due to different levels of attention to grammar. It is important, though, that in both experiments, learners patterned with the natives.

In conclusion, after the early ERP studies showing clear divergence between native and bilingual processing mechanisms of the inflectional morphology, later studies with improved designs attested to a higher level of convergence. It is not the case that the L2 learners, even at advanced proficiency, always perform as native speakers. However, factors such as more arduous processing and a lower degree of automaticity explain a lot of the differences. Importantly, acquisition of the inflectional morphology may require effort, and the knowledge may be hard to implement in real language processing, but neither nativelike acquisition nor nativelike processing are impossible.

12.4 Parsing and syntactic processing in the L2

We are going to look next at syntactic processing. Of course, processing of the inflectional morphology, discussed in the previous section, is truly syntactic processing, since all syntactic features are captured in inflectional morphology. In this section, we will pay attention to the visible reflexes of syntactic processes: movement of phrases at a distance and their interpretation.

But first, let me make abundantly clear what I assume to be the division of labor between the grammar and the parser (the language processing system), since this is a controversial but very important issue. The parser is a grammatical analyzer which quickly checks the words of the incoming sentence, considers their morphosyntactic information, and integrates them into syntactic structure in an incremental way (Crocker 1996, Fodor 1998). Based on the structure that the parser builds, a partial semantic interpretation is computed at each stage of syntactic derivation (see Chapter 2). Interpretation building takes into account the semantic information from the words, as well as pragmatic inferences and information from the context of the utterance. All this takes place in mere milliseconds,

that is, as fast as lightning. Processing linguistic input in real time reveals information about the way representations of linguistic knowledge are structured and stored in the mind/brain of the hearer/reader. Without much argument, I have assumed here a view of linguistic processing which postulates that there is a *considerable overlap* between the grammar and the parser (following the seminal work of Janet Fodor, Lyn Frazier, and many others), and that the parser's operations are based on grammatical representations.⁶ As we saw in Chapter 3, the grammar proposes (alternative construals), the parser disposes (which one should be selected as the correct one). There are ample research findings to support this assumption.

Teaching relevance

The teaching relevance of this assumption that the grammar and parsing go hand in hand is clear: the more practice learners have at comprehending and producing the second language, the more their underlying grammar will grow, hand in hand with their processing skills. An active parser can aid a developing grammar.

Let us take a more detailed look at some theoretical proposals for native syntactic processing. A model of native language processing called Late Assignment of Syntax Theory (LAST), Townsend and Bever (2001), proposes that all syntactic processing advances along two lines in parallel: a “pseudosyntactic” analysis based on canonical templates and semantic roles, which is immediately linked to semantic representations; and a true syntactic analysis that takes place independently of the semantics. The pseudosyntactic analysis is a “quick and dirty” parse dependent on associations and frequencies in the input. It moves the *wh*-words back to their original argument positions and then applies canonical template strategies (such as Agent–Verb–Theme)⁷ to provide an initial meaning analysis. The

⁶ For example, take this citation from Duffield (2006: 57). No one can say:

that a particular piece of linguistic performance—whether it comes from traditional behavioral methods such as response latencies, or from more modern techniques such as event-related potentials—is uniquely due to the grammar or to the processing system. [...] grammatical competence is always mediated by the processing system (in virtue of being part of it)...

⁷ This is what the template does. If the parser assumes that every incoming sentence containing NP+VP+NP (e.g., *George offers help*) exemplifies the Agent–Verb–Theme

syntactic analysis proper goes through a cyclical derivation for the whole clause and offers a completely specified syntactic representation. At the end of the two parallel processes, their outputs are compared and if there is a match, the computed meaning is stored. If there is no match (due to garden path, reanalysis, etc.), the computation can start again at various points. The two lines of computation are argued to be independent up to the comparison point. Ferreira's Good Enough model for initial representations (Ferreira, Bailey, and Ferraro 2002) is reminiscent of the LAST model, although it may differ in the details.

Clahsen and Felser's (2006) Shallow Structure Hypothesis extends the insights from these two models to second language processing. The hypothesis postulates that the pseudosyntactic, or good-enough, way of parsing the incoming input is *the only way* bilinguals can parse language. According to the Shallow Structure Hypothesis, it is not just a processing issue: the L2 grammar does not provide the kind of information required to process complex syntax in nativelike ways, forcing L2 learners to fall back on "shallow" parsing strategies. L2 learners use lexical-semantic and pragmatic information as well as templates of predicate argument structure (Agent–Verb–Theme) to interpret incoming strings of words in a minimal (shallow) semantic representation, without mapping detailed and complete syntactic representations onto semantic representations. Several sentence-processing studies by Clahsen and associates have found that L2 learners indeed rely less on hierarchical phrase structure or on abstract syntactic elements such as movement traces, than native speakers during comprehension. Direct evidence for the SSH comes from the online processing of filler–gap dependencies. For example, as indicated by Marinis et al. (2005), learners do not process the intermediate copies of the moved *wh*-word *who* in examples like (12) as native speakers do.

- (12) [DP The nurse [RC [**who**] the doctor argued [CP [~~who~~₂] that the rude patient had angered [~~who~~₁]]]] . . . is refusing to work late.

The relative pronoun *who* starts out as the object of the verb *to anger* and on its way to the top of the relative clause RC, it stops over in the

mapping between arguments and theta roles, it will be over 85% correct. Probabilistically speaking, that's a safe assumption. Sentences such as *The key opened the door*, not following the template, are rare in English.

intermediate CP position in front of *that*. In example (11), these two positions are marked ~~who~~₁ and ~~who~~₂. When the relative pronoun stops in the ~~who~~₂ position, it leaves a copy of itself and continues on its way. In the generative theory of *wh*-movement, copies of movement have psychological reality although they are not pronounced, and they are indicative of abstract structural representations of sentences. Since learners in the Marinis et al. (2005) study did not show sensitivity to this intermediate copy, as indicated by reading times, the researchers argued that second language speakers rely on lexical–semantic and predicate–argument relations between words, when they are processing such long-distance dependencies. In other words, their processing is meaning-based, not structure-based. Furthermore, Clahsen and Felser interpreted this to indicate that “L2 processing is different because of inadequacies of the L2 grammar” (Clahsen and Felser, 2006: 120). Felser and Roberts (2007) present more evidence for L2 speakers using quantitatively different processing routines from the native speakers.

Opponents of the Shallow Structure Hypothesis try to show that L2 speakers are indeed sensitive to such structural representations that are needed to calculate sentence meaning online. Among many recent studies, Omaki and Schultz (2011) demonstrate that both their English native speakers and Spanish learners of English obey relative clause island constraints. Aldwayan, Fiorentino, and Gabriele (2010) also show evidence that Najdi Arabic speakers process English *wh*-movement actively guided by syntactic constraints. Dekydtspotter and Miller (2013) study the activation of intermediate traces of *wh*-movement in a priming experiment and argue that their results are better explained by weak activation of semantic concepts, probably due to lexical access difficulties. They caution that research on the processing of *wh*-dependencies in sentence processing must give full consideration to lexical activation mechanisms. Dekydtspotter et al. (2012) report reading time asymmetries in L2 speakers that are consistent with observing constraints on binding. Within ERP studies, Reichle and Birdsong (2014) tested sensitivity to contrastive focus in French among advanced classroom learners. They observed a nativelike negativity in their more proficient learners.

Apart from findings inconsistent with it, one line of argumentation against the shallow processing idea for bilinguals, due to Indefrey (2006), counters that a number of native speakers also resort to using semantic-based processing most of the time. These happen to be low-educated, low-reading-span or nonproficient native speakers (Pakulak and Neville 2010).

The effect of lack of experience with a certain construction can be the same for monolingual native speakers and for bilinguals in their second language, as Dąbrowska's work on the passive has shown (although with a comprehension task). See more on this in the section on individual differences. The point that a lot of researchers make is that shallow processing is not exclusive to bilinguals.

Another interesting factor that may modulate shallow processing is learners' type of exposure: naturalistic (in the country where the L2 is spoken) versus in the classroom. In the typical case, naturalistic exposure, especially study abroad, would furnish richer and more varied language experience. Two recent experimental studies are relevant here. Pliatsikas and Marinis (2013a) tested Greek learners with either naturalistic and/or classroom exposure to English on the past tense rule. Both their highly proficient learners with an AoA (age of arrival) of 8–9 years performed similarly to natives, therefore suggesting that the type of exposure is not a factor in automatic processing of morphosyntax while age of acquisition and proficiency are. Pliatsikas and Marinis (2013b), however, examined the processing of two similar groups of Greek–English bilinguals with the same age of acquisition (8–9 years) and again with naturalistic or predominantly classroom exposure. Using the same stimuli from Marinis et al. (2005) exemplified in (11), they found that their naturalistic learners (but not the classroom learners) were indeed processing the intermediate traces like native speakers. These latter results suggest that linguistic immersion can indeed lead to nativelike abstract syntactic processing in the L2. They are also in line with the importance of experience for nativelike and efficient parsing.

On balance, the SSH appears to be falsified by numerous recent studies using self-paced reading, eye tracking, and ERP components. A lot of the studies reviewed here demonstrate that learners are sensitive to syntactic constraints in their second language. Apparent divergence may be due to linguistic experience with a specific construction, proficiency, and type of exposure.

12.5 Integration of meaning in syntactic processing

Most current models of language processing agree that, in online sentence processing, different types of constraints (lexical, morphological, syntactic

and semantic) are very quickly taken into consideration during speaking and listening (or reading). Constraints on how words can be combined structurally operate alongside qualitatively distinct constraints on the combination of word meanings, on the grouping of words into phonological and intonational phrases, and on their integration into discourse. There is evidence that semantic, syntactic, and phonological unification processes (combinatory processes) all operate at the same time and influence each other to some extent. An example of semantic unification is the integration of word meaning into an unfolding discourse representation of the preceding context. For instance, the majority of common English words have more than one meaning. In the interaction with the preceding sentence and/or discourse context, the appropriate meaning is selected, so that a coherent interpretation results.

In this section, we will examine the evidence on incorporating lexical-semantic and subcategorization information into the online computation of meaning. To anticipate the findings, it turns out that L2 speakers are generally very good at integrating semantic information into the parse.

The crucial sentences where this is studied are garden path sentences. In the processing of garden paths, the subcategorization of the verbs involved is of utmost importance. But first, let's back off a little and revise what speakers know when they know a verb: quite a lot, actually. Each verb is learned with information about which thematic roles (theta roles) are needed for the sentence with this collocation to be acceptable (see Chapter 3, Section 3.4 and note 7). This information is known as verb valency (as in chemistry) or subcategorization: *give* takes an Agent, a Theme, and a Goal (*John gave a package to Mary*); *put* takes an Agent, a Theme, and a Location (*John put the apple on his head*), etc. Theta roles are mapped onto syntactic arguments (subject, direct object, indirect or prepositional object, etc.), not always in a uniform fashion. In a theory of incremental parsing based on theta roles and early satisfaction of grammatical principles (Pritchett 1992), the parser will use several sources of syntactic information. Let us look at what happens in the incremental processing of (13).

(13) John gave a package to Mary.

After *John* appears in reading or in speech, the parser expects a verb. This verb, if finite, will satisfy the syntactic requirement of checking the case of the subject (Nominative). It will also assign the Agent theta role to *John*.

The verb *give* being tri-valent, that is, needing a Theme and a Goal in addition to the Agent, the parser will expect appropriate noun phrases or prepositional phrases to appear next. If these predictions are fulfilled, the parse can terminate successfully, producing an interpretation. Processing breakdown can occur if the words after the verb are not as expected by the parser for the immediate satisfaction of subcategorization and syntactic requirements. Furthermore, the parser can be fooled into positing one structure (temporarily): the appearance of subsequent words indicate that another structure is really warranted, so reanalysis is required. Take the garden path (GP) sentence in (14a).

- (14) a. After Bill drank the water proved to be poisonous.
b. Sam warned the student cheated on the exam.
c. After Sam arrived the guests began to eat.

Speakers know that the verb *drink* can appear with or without a Theme argument in object position. In the incremental appearance of the sentence words on the screen (in a self-paced reading task), *the water* satisfies grammatical and semantic requirements, so the temporary structure *After Bill drank the water* can be plausibly posited. After the appearance of the following verb, *proved*, it becomes clear that this new verb will also need a subject argument (although not an Agent). *The water* is reanalyzed as that argument. Note that on top of taking time and effort, reanalysis depends on the appropriate lexico-semantic and syntactic knowledge being available online.

The groundbreaking research on how L2 learners integrate this knowledge into the comprehension of a sentence is due to Juffs and Harrington (1996) and Juffs (1998a,b). The researchers examined the processing of sentences as in (14a,b,c) and found that Chinese–English bilinguals were slower to read the disambiguating verbs (*proved*, *cheated* in (14a,b)) in garden path sentences, as compared to non-GP sentences containing intransitive verbs (e.g., *arrive* in (14c)). This shows that learners were sensitive to the fact that *drink* and *warn* can be both transitive and intransitive, in English, but *arrive* is only intransitive. In a later experiment, Juffs (1998a) compared the processing of similar garden path structures by bilingual speakers with Chinese, Japanese, Korean, and Romance as native languages. As Figure 12.4 illustrates, the bilinguals are slower than the English native speakers, but their processing of garden paths is uniform: all groups are

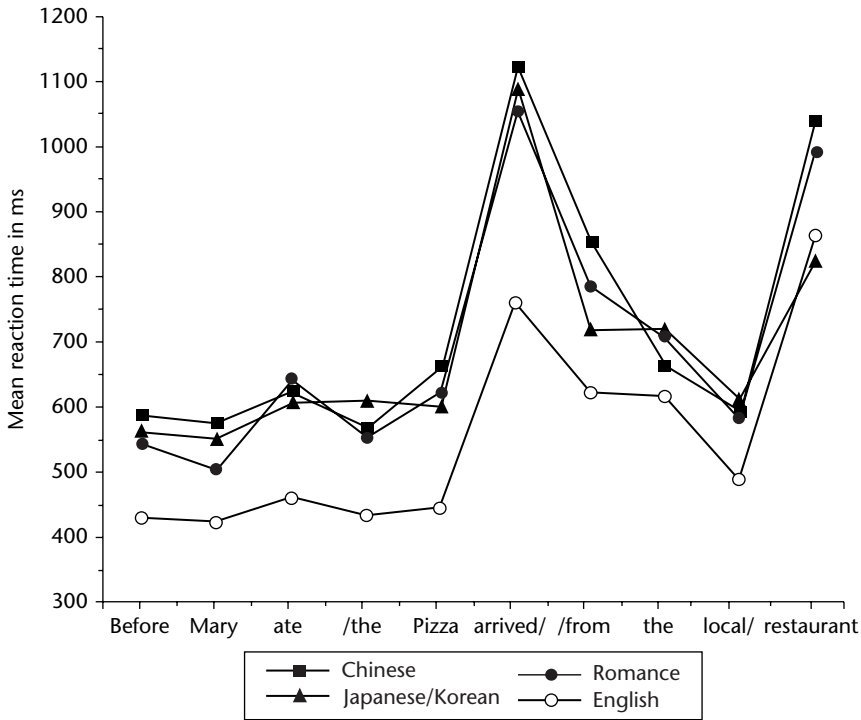


Figure 12.4 Meaning reading times for garden path sentences by four groups of speakers, from Juffs (1998a)

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surprised by the appearance of the verb *arrived*, and take time to reanalyze *the pizza*.⁸

In a similar vein, Dussias and Cramer Scaltz (2008) explored sensitivity to verb biases, the information on what type of argument (NP or complement clause) follows a verb most often. In a self-paced reading experiment, they found that Spanish–English bilinguals’ reading times were longer when the continuation went against the verb biases in English.

- (15) a. The CIA director confirmed the rumor could mean a security risk.
b. The ticket agent admitted the mistake when he got caught.

⁸ See Juffs (1998a) for the findings on the non-GP sentences. It is on those sentences that the performance of the learners groups diverged, suggesting L1 influence.

Learners were sensitive to verb biases when they overlapped in the L1 and L2. When the Spanish verb biases were contrary to English, though, the learners transferred their native biases. Results of these and other experiments, taken together, demonstrate that bilinguals process the incoming linguistic signal incrementally and try to accommodate each word subcategorization information and bias, as long as they know it.

Roberts and Felser (2011) explored whether L2 speakers use plausibility information in their incremental processing. Even if a verb is known to be predominantly transitive, such as *read*, some objects following *read* are more plausible than others. The researchers used this to see how much plausibility will influence the ease of recovery after garden path sentences.

- (16) a. The journalist wrote the book had amazed all the judges.
b. The journalist wrote the girl had amazed all the judges.

Both sentences require reanalysis, but in (16a) the initial analysis is plausible (*the journalist wrote the book*) while in (16b) it is not (*the journalist wrote the girl*). The researchers hypothesized that readers will have a harder time recovering from the initial wrong analysis if it was plausible, that is, in cases such as (16a), which would be indexed by longer reading times at the disambiguating complement clause. That's exactly what they found, for natives and L2 speakers alike. They argued that the bilinguals' commitment to a plausible initial analysis is stronger than to an implausible one, and thus harder to abandon in the face of new evidence. However, this was not the case in constructions with fronted adjunct clauses, as in (17).

- (17) a. While the band played the song pleased all the customers.
b. While the band played the beer pleased all the customers.

The L2 readers continued to show elevated reanalysis reading times no matter whether the initial analysis was plausible or not. The authors argued that the more complex computation of the adjunct clause had affected the readers' ability to process incrementally. Still, L2 readers showed they could do lexico-semantic integration, even if processing complexity affected them negatively.

In sum, numerous analyses and comparative studies of bilingual processing acknowledge that lexico-semantic processing is not as challenging to learners as grammar processing. An indication of this is the fact that

learners are quite accurate integrating lexico-semantic knowledge at lower levels of proficiency, in comparison to morphosyntactic knowledge. In contrast, the differences observed between native and L2 speakers lie in the area of grammar processing. For instance, L2 learners find it harder to recover from mis-analysis in some garden path sentences, and have trouble establishing grammatical relationships across clause boundaries, or when the material to be processed is structurally more complex. This state of affairs does not mean that morphosyntactic processing is not accessible, or that their linguistic representations are faulty; it just means that grammar processing is harder and nativelike behavior patterns are attained at later stages of development.

12.6 Individual differences in grammar processing

Language processing is a very complex process that involves a rapid application of lexico-semantic and morphosyntactic knowledge, the resulting parse being evaluated in the light of discourse pragmatic and world knowledge. In addition, during the structural analysis of the signal, predictions are generated as to what might be expected further along in the sentence string. Not surprisingly, researchers have established that there is a lot of variability among native speakers in all of these processes. Research so far has established that working memory, as tested by memory span tasks, shows some correlation with successful and fast processing of complex and/or long strings. It makes sense that individuals who can retain a larger number of words or numbers in short-term memory would also be able to use that memory to store and manipulate more complex linguistic structures. For example, King and Just (1991) found that participants with low scores on a test of verbal working memory ability produced longer reaction times on difficult regions of test sentences, compared to high-span participants. They were also less accurate on related comprehension questions. The idea was that taxing the lower memory resources with a complex computation led to more errors in comprehension.

However, Caplan and Waters (1999) argued against the existence of a single pool of working memory resources responsible for language comprehension and in favor of a multi-resource theory. Recent reviews of the available research (Farmer, Misiak, and Christianson 2012) suggest that verbal memory explains a rather small amount of variance among natives.

Instead, Farmer et al. (2012) propose an experience-based hypothesis: past linguistic experience substantially influences sentence processing. The more experience an individual has with a specific construction such as the passive, or in reading long and complicated sentences, the easier that processing becomes. (The teaching relevance of this thesis is quite clear, right?) In addition, differences in cognitive control, as measured by stroop tasks⁹ and the like, may be affecting processing of ambiguous sentences (e.g., *Jane hit the man with the umbrella*) and garden paths, because one needs cognitive control in order to direct attention away from one analysis and contemplate another analysis (Novick et al. 2005).

In a recent review, Phillips and Ehrenhofer (in press) argue that the single substantial difference between native adult and native child processing is that children are not good at reanalysis (as needed in garden paths) or calculation of two meanings in ambiguous sentences. Children's first analysis is their last analysis, they argue, being their one and only analysis. We have already seen that in adult native speakers, this is not the case. Phillips and Ehrenhofer (in press: 7) plausibly attribute this child–adult difference to children's limitations in cognitive control abilities, and not to a growing of the grammar (Novick et al. 2005, Mazuka et al. 2009). For example, “children show broad difficulties in revising initial action plans, likely due to the delayed maturation of the frontal lobe, the primary home of executive function abilities (Davidson et al. 2006).” Language difficulties are part of a broader maturation picture.

Now, if this interpretation is on the right track and children have difficulties parsing some complex syntactic structures due to executive function limitations, could this explanation work for L2 learners? In principle, it should. Phillips and Ehrenhofer (in press) say no, but I disagree. Executive function in bilinguals is taxed by the constant inhibition of the language not in use and by the juggling of two lexical items for each concept, different grammatical rules, etc. Individual differences in cognitive abilities (e.g., executive function) should, again in principle, have a higher impact on processing in bilinguals than in monolinguals. What could the evidence be in support of such a processing explanation of L1–L2 differences?

⁹ A stroop task is a demonstration of interference in task reaction. When the name of a color (e.g., “blue”) is printed in a color different from the name (e.g., the word “red” printed in blue ink instead of red ink), naming the color of the word takes longer and is more prone to errors than when the color of the ink matches the name of the color.

We have established in the previous two sections that second language learners process the language in a similar, even if not in an identical, fashion with native speakers. They have more difficulty with complex structures (sentences such as (12) above), as well as structures where they have to keep a lot of material in short-term memory (sentences such as (17)). It is logical to expect, then, that at least some of the divergence between natives and bilinguals may be due to individual memory and cognitive differences. In addition, many researchers of second language acquisition suggest that native speakers also exhibit a range of processing capabilities, as do L2 learners, so we have to be careful in choosing comparison groups. In this section, we will briefly summarize what is currently known on this topic, provided that research on individual differences in bilingual processing is in its infancy.¹⁰ However, it is an area of very vigorously developing research, and the reader can expect to hear much more about it in the future.

The most important factor impacting processing that L2 researchers test is L2 proficiency. Note that in this way they are applying Farmer et al.'s (2012) suggestion that linguistic experience may influence processing more than memory span. Intuitively it makes sense that L2 learners of higher proficiency are also those with more experience in that language. However, the findings are more complicated. Overall, it is attested that higher proficiency learners pattern as native speakers only when their attention is drawn to grammatical form (not meaning) in consideration of an additional experimental task. For example, Jackson and van Hell (2011) asked Dutch learners of English to process sentences such as those in (18).

- (18) a. Who do you think ____ met the tourists in front of the museum?
(subject *wh*- word extraction)
- b. Who do you think the tourists met ____ in front of the museum
(object *wh*- word extraction)

Learners had to read as well as give grammaticality judgments on acceptable and unacceptable sentences of this sort. The less proficient learners experienced processing breakdown on the subject extraction sentences,¹¹

¹⁰ See Juffs and Harrington (2011) for an overview of the literature on working memory, and Roberts (2012) for a review of individual differences in L2A.

¹¹ Subject extractions are considered to be computationally more complex than object extractions, and they are dispreferred by children and adults alike.

while the more proficient learners patterned with the native speakers. Comparing the results of two self-paced reading studies makes this point abundantly clear. Jackson and Dussias (2009) asked English learners of German to read and to judge the grammaticality of sentences as in (19).

- (19) a. Wer denkst du, bewunderte den Sportler nach den Spiel?
Who.NOM think you admired the.ACC athlete after the game
‘Who do you think admired the athlete after the game?’
b. Wen denkst du, bewunderte der Sportler nach den Spiel?
Who.ACC think you admired the.NOM athlete after the game
‘Who do you think the athlete admired after the game?’

In that study, advanced learners showed nativelike processing of the questions. In subject extractions (19a), they tried to integrate the nominative-marked *wh*-word into the parsed sentence as soon as possible, that is, in matrix subject position. When they encountered *du* ‘you’ as the subject of the matrix clause, they experienced difficulty, reflected in longer reading times for that segment. This was not the case for object extraction sentences as in (19b). Upon reaching the complement clause, however, reading time preferences reversed for both the native and the L2 German speakers, with longer reading times on object extractions compared to subject extractions. This was attributed to the fact that the object extraction sentences were more difficult to process because the fronted object had to be kept in working memory for a longer time. In a continuation of the first study that had subjects read without grammatical evaluation (Jackson and Bobb 2009), no reading time differences were observed. There were no differences in the complement clauses of the two conditions (subject and object extraction). Crucially, no effects of proficiency or working memory capacity were attested.

The comparative findings of the two studies suggest that bilinguals process better when their attention to grammar is heightened. They often adopt processing strategies to fit the demands imposed by the task, especially when challenged by complex processing. The authors write that

these resulting strategies may not mirror the strategies employed by a majority of native speakers of the language in question, and they may be highly individualized, varying dramatically from L2 speaker to L2 speaker. Nevertheless, they highlight the ability of L2 speakers to make sophisticated use of the linguistic and cognitive resources they have at their disposal to successfully process and comprehend L2 input. (Jackson and Bobb, 2009: 631.)

I think the connection with executive function here is clear, although completely speculative: in a nutshell, when they put their mind to it, L2 speakers are successful even with the most complex structures. However, much more research on this idea is needed.

Another individual factor that can account for L1–L2 differences is processing efficiency. It is well known that L2 readers are slower than native readers, especially if they are reading in a script not their own.¹² However, as Roberts (2012) points out, the term “processing efficiency” is at this time not well defined.

It could relate to efficiency in a number of different processes undertaken during language comprehension, from decoding of orthography/speech sounds, to lexical access and selection, to integration of syntactic and other knowledge, as well as to prediction or anticipation of up-and-coming input. (Roberts 2012: 181.)

Roberts and Felser (2011) argue that being slower does appear to affect parsing procedures, particularly when the task involves reading for comprehension only. They also align reading speed with proficiency.

In a study particularly designed to test the effect of reading speed, Kaan, Ballantyne, and Wijnen (in press) used a self-paced reading task with English native speakers and Dutch learners. When they performed the statistical analysis of the grammatical manipulations (subject–verb agreement in object relative clauses) not controlling for reading speed, that is, including all of the participants, the L2 group performance appeared to differ from the native group in that the effect started later and was smaller than in native speakers. However, when the authors selected a group of L2 speakers who matched the natives in reading speed, differences in processing patterns disappeared. In other words, language background did not have any effect above and beyond reading speed.¹³ In addition, the readers in this study had to perform an end-of-sentence verification task with True/False answers. Only the faster L2 readers but not the slower ones showed a nativelike pattern in this task. The researchers interpret the findings to suggest that L2 learners and native speakers have the same grammar and are able to build the same syntactic structures during online processing, but

¹² Some recent exceptions are Hopp (2015) and Kaan, Ballantyne, and Wijnen (in press), who found that some of their bilingual participants read faster than their native readers.

¹³ Note that in this study, in order to create reading speed-matched groups, some of the fastest L2 readers had to be excluded, as they read faster than native speakers.

may differ in terms of allocation of processing resources (see also Hopp 2010, McDonald 2006). Even if some L2 speakers read at the same speed as the native speakers, they likely have more difficulty computing and maintaining the syntactic representations in short-term memory. This divergence may be due to slower lexical access, lower activation of the relevant information, or fewer available resources (e.g., because bilinguals need to inhibit competing native representations).

The interplay between processing efficiency and speed is addressed in Hopp (2010). It is generally assumed that when readers are forced to speed up their reading, their processing load increases. Hopp used a moving window word-by-word presentation, but varied the speed of presentation from 250 ms plus 17ms per each letter (Speed 1) to 155 ms (Speed 2), 105 ms (Speed 3), 88 ms (Speed 4), and 71 ms (Speed 5). Four groups of native speakers were tested on Speeds 2 to 5. The results showed that under conditions of speeded grammaticality judgments, the accuracy of native speakers deteriorated to the level of L2 learners at Speed 1. However, that happened only for gender violations, not for subject–verb agreement violations. The different treatment of gender and number violations fits in with behavioral findings determining that gender violations are particularly problematic for L2 learners, even if they have grammatical gender in their native language (e.g., Franceschina 2005). Thus, native speakers' performance under increased processing load and L2 speakers' performance converge on selected violations (e.g., for gender but not number). What is difficult for nonnatives is also difficult for natives under conditions of forced, less efficient processing. This finding suggests that even if natives and bilinguals share grammatical knowledge and processing architecture, bilinguals may not be able process the L2 as efficiently as natives.

Finally, Hopp's (2015) study brings forward another interesting dimension along which L2 learners may differ: individual allocation of resources, or attention, to either form or meaning. Hopp discusses L1 German–L2 English learners' use of morphosyntactic, semantic, and subcategorization information in the processing of temporarily ambiguous sentences as below.

- (20) a. When the girl was praying, the boy made some funny noises.
(Control)
b. When the girl was praying the boy made some funny noises.
(Intransitive)
c. When the girl was playing he made some funny noises. (Pronoun)

- d. When the girl was playing the boy made some funny noises.
(Implausible)
- e. When the girl was playing the piano made some funny noises.
(Plausible)

In order to gauge the individual factors of proficiency, reading speed, working memory span, and allocation of processing resources, Hopp tested the participants in a lexical decision task, a reading-span task, and a word-monitoring task. The main task was eye tracking during reading. While he confirmed previous findings of successful lexico-semantic integration, a plausibility effect, and morphosyntactic difficulties, the more interesting findings relate to individual processing profiles. The researcher divided the learners into low integration, mid integration, and high integration groups, depending on the syntactic integration ability they showed in the word-monitoring task. The low integration group was revealed as using plausibility and subcategorization information for recovery from garden paths but did not use morphosyntactic information. On the other hand, the high integration group showed a fully nativelike processing pattern in that it recruited morphosyntactic as well as subcategorization information incrementally in first-pass and second-pass measures of processing.

It seems then that individuals (bilingual learners as well as monolingual natives) might differ in the attention they pay to either form or meaning. This division is reminiscent of the dual pathway described by Townsend and Bever (2001) and Ferreira and colleagues (2002) for native processing (Section 12.4). Recall that according to these accounts, sentence processing can proceed along two routes: (a) a shallow analysis that relies on frequent and predictable surface patterns including lexical-semantic and plausibility information; and (b) a deep, syntactic parse that exploits grammatical structure. In natives, these routes are argued to work in parallel, but wide individual differences exist, too (Pakulak and Neville 2010). In the L2 learners participating in Hopp's study, accuracy did not seem to correlate with proficiency, lower working memory, or problems in lower level linguistic processing. The author suggests the different performance of individual learners on his tests stems from their need to allocate resources efficiently and to these resources being inherently limited in bilinguals.¹⁴

¹⁴ See also Wilson, Sorace, and Keller (2009).

Furthermore, hitherto unexplained variation among speakers may also play a role here. The recent findings of Tanner and van Hell (2014),¹⁵ an ERP study of monolingual native speakers detecting morphosyntactic violations, are illuminating. Recall that morphosyntactic violations are assumed to elicit a biphasic LAN–P600 effect, but not an N400 effect. The study did find such a biphasic pattern of responses in the group results. However, they did not find the expected biphasic LAN–P600 response for most of their individual participants. Instead, individuals’ brain responses varied along a continuum between an N400 and a P600, with some biphasic patterns in between. The researchers explicitly argued that the individual patterns of responses to morphosyntactic violations were very similar to those uncovered in some bilingual speakers. In other words, we may be seeing variation among L2 learners because variation exists among ALL speakers of language, irrespective of nativeness, proficiency level, and education profiles. This is an area of research that will bring important new discoveries relevant to SLA in the future.

The studies we reviewed in this section all point to the conclusion that examination of individual differences should be an obligatory element of future psycholinguistic studies. L2 learners, just like native speakers are NOT a homogeneous group. Statements amounting to “L2 speakers do this or that” should not be acceptable if they do not differentiate between individual performances based on L2 proficiency, memory capacity, speed of reading, and processing efficiency.

12.7 Conclusion

Processing differences between monolingual native speakers and bilingual or multilingual L2 users, together with reliable and copious input, is emerging to be the most powerful explanation of L1–L2 competence differences. Researchers who would not agree on much else in L2A would nevertheless agree that bilingual processing is among the most important areas of L2A. Slower or more belabored processing in L2 learners should be distinguished from competence difference explanations such as the Fundamental Difference Hypothesis (Bley-Vroman 1989, 2009) and the Representational

¹⁵ See also Tanner et al. (2013).

Deficit view (Hawkins and Hattori 2006, Tsimpli and Dimitrakopoulou 2007). In other words, an L2 user may appear to be different from a native language user for two reasons: either because their processing is different or because their competence (hence, linguistic representations) *and* processing are different. We reviewed accounts of L2 processing divided into two main positions, largely as follows. One position, exemplified in the work of Clahsen, Felser, Paradis, and Ullman, argues that the L2 processing mechanisms are qualitatively different. Furthermore, Clahsen and Felser attribute the processing divergence to an underlying competence divergence. The other position maintains that processing mechanisms in the second language are essentially the same as in the first language, but the pressures of bilingualism can lead to apparent L1–L2 differences. Processing preferences and routines can transfer from the native language, but can also be overcome. This position is exemplified in the work of Dekydtspotter, Dussias, Juffs, Gabriele, Omaki, Schulz, VanPatten, and many others. For recent reviews of this position, see Dekydtspotter (2009) and Belikova and White (2009). In the future, we can expect a lot of the answers to the fundamental questions of SLA to come from, or partially from, processing findings.

12.8 Exercises

Exercise 12.1. Do you agree with the logic that the CPH should be falsified if we find even one or two late L2 learners who are completely nativelike? In this chapter we mentioned that if the strong CPH claim is to be supported with neurophysiological findings, studies must show that successful learners use *different brain mechanisms* than native speakers. Discuss what that difference might mean in actual findings, keeping in mind the studies exemplified in the chapter. Is it enough to show that L2 and L1 speakers exhibit different behavioral measures (be that reaction times, eye movement saccades, ERP reflexes, etc.)? How about *different processing mechanisms*?

Exercise 12.2. Sato (2007) is a dissertation study, summarized in Clahsen et al. (2010), investigating L2 learners' sensitivity to subject–verb number agreement and pronominal case. In order to assess the possible influence of the presence versus absence of these grammatical categories in the learners' L1, Sato examined three groups of learners from typologically different L1 backgrounds (German, Japanese, and Chinese) in a series of speeded

grammaticality judgment tasks. Note that German is similar to English in that it marks both case and S–V agreement, Japanese has a morphological case but lacks S–V agreement marking, whereas Chinese lacks both. All L2 learners scored 70% or above in the Oxford Placement Test, placing them within the intermediate to advanced proficiency range. They demonstrated close to ceiling knowledge on case and agreement in a separate offline task, confirming that they were aware of the relevant grammatical distinctions.

In the speeded judgment task, learners saw simple three-word sentences such as *We regularly sneezes* and *He admires she*. All sentences became ungrammatical at the final word. The experimental sentences were presented word-by-word at a rate of 350 ms per word to increase processing pressure. Participants were asked to judge whether a given sentence was well formed and meaningful immediately after reading the final word. Table 12.1 summarizes the results.

Question 1. Are learners more accurate on case or on agreement? How about the natives?

Question 2. Are learners faster on judging case or judging agreement? How about the natives?

Question 3. The results presented in Table 12.1 are based on the accuracy and speed of answers of ungrammatical sentences only. Why do you think the researcher made this decision?

Question 4. Is there L1 influence in these data? Why or why not? How does that square with the Failed Functional Features Hypothesis (Hawkins and Casillas 2008), according to which formal features are only accessible to adult L2 learners if they are instantiated in their L1, but not if they are not present in their L1?

Table 12.1 Summary of experimental findings on English case and agreement inflections: Error rate and RT differences between the ungrammatical Agreement and the ungrammatical Case conditions

	L1	German L2	Japanese L2	Chinese L2
Errors (Agreement–Case)	–0.7%	4.4% (*)	9.7%*	17.2%*
Response times (Agreement–Case)	3 ms	162 ms*	298 ms*	527 ms*

Note: Source data from Sato (2007).

* Significant at $p < .05$ by subjects and items.

(*) Significant at $p < .05$ by items.

Question 5. Here are some factors that could have affected the differential difficulty:

- a. phonological salience (DeKeyser 2005)
- b. slower processing speed (McDonald 2006)
- c. computational resource limitations (McDonald 2006)
- d. grammatical complexity in terms of number of formal features instantiated in the morphology
- e. S–V agreement dependencies span the entire clause (and thus require comparatively complex structural scaffolding), whereas the object case is assigned locally within the verb phrase.

Discuss these factors in turn and decide which one could explain the findings.

Exercise 12.3. Find and download the following study: A. Foucart and C. Frenck-Mestre (2011). Can late L2 learners acquire new grammatical features? Evidence from ERPs and eye-tracking. *Journal of Memory and Language* 66, 226–248. Read it while thinking of the following questions. Then try to answer them without being influenced by the interpretation of the authors (where appropriate).

Question 1. From the introduction, make a list of the factors affecting acquisition of grammatical gender and its processing online.

Question 2. Are the research questions appropriate?

Question 3. In French, adjectives can be post-nominal, the more frequent option, and pre-nominal, the less frequent option. How is frequency used in this experiment? What other factor does it interact with?

Question 4. What is the rationale for the third ERP experiment?

Question 5. What does eye tracking in Experiment 4 add to the mix of experiments?

Question 6. What L2 processing theories are addressed in the general discussion? Which ones are the findings compatible with?

Question 7. Grammatical gender is a new feature for these learners, but out of the three possible placements of adjectives (pre-nominal, post-nominal and predicative) only the post-nominal placement is new for learners of French. How does this fact affect the accuracy of the learners? Which placement are they most accurate on?

Question 8: Which theoretical proposal does this finding support? Could there be other reasons for this accuracy, such as frequency of the collocation and instruction?

Exercise 12.4. Testing for individual differences. Imagine that you are in an ideal world and you can test everything and everyone you want with whatever techniques you deem appropriate. (Wouldn't that be great.) You are given the task to identify how individual differences affect L2 processing.

Question 1: What kind of research design would you use?

Question 2: What factors of difference between individuals would you include?

Question 3: What linguistic property or properties do you think are good to test in such a design? Will the properties be different in the L1 or L2, or the same? Why?

Question 4: Make a table of potential findings and identify what findings would suggest that individual differences among monolingual native speakers are smaller than, or similar to, those among L2 speakers.